

ChE 210 MATERIAL AND ENERGY BALANCES
Spring 2020

- 01/20 No Lecture (Martin Luther King, Jr. Holiday)
- 02/21 First Midterm Exam
- 03/23-27 No Lectures (Spring Break)
- 04/10 Second Midterm Exam
- 05/05 FINAL EXAM 10:30 am - 12:30 pm

LECTURER:

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TEXTBOOK: *Introduction to Chemical Processes: Principles, Analysis, Synthesis*,
Murphy, 2007

EXAMS, HOMEWORK, AND GRADING:

There will be two midterm exams, several unscheduled quizzes, and one final exam. All exams are required. There will be three 50 minute-lectures per week typically (Mon-Wed-Fri) and one "discussion session" (Tues), although this schedule may change occasionally. During the discussion sessions, homework problems will be worked, numerical methods introduced, and other new material covered. The grader is under no obligation to grade homework that is late, illegible, or sloppy. Not scoring at least half of the total homework grade is grounds for failing the course.

The grades in the course will be determined as follows:

Discussion	5%
Homework	10%
Quizzes	15%
Exams (20% each)	40%
Final Exam	30%

Please note that the final exam is on Tuesday May 5, 2019 from 10:30 am - 12:30 pm and that there will normally be no special arrangements for the exam to be taken at a different time. (Final exam conflicts will be resolved according to https://registrar.uic.edu/current_students/calendars/final_exam_schedule.html#complete-schedule)

AIMS, SCOPE, LEVEL

This course is a continuation of the material covered in ChE 201 Thermodynamics. In that previous course, you became familiar with the concepts of energy and entropy balances for pure components systems. Also, we will make use of the material covered in ChE 205 for solving problems. In ChE 210, we will study more complex systems including mixtures and reacting systems. We will also include more complex processes with several feed and effluent streams. The chemical processes will also contain unit operations such as separators and reactor. This increased level of complexity will require more sophisticated methods of analysis. We will also begin to rely more on numerical methods to help us solve these more challenging problems. You will become familiar with some basic numerical methods and applications.

This course lays out the general approach of chemical engineering. Later courses will deal in more detail with many of the particulars discussed in ChE 210. For instance, Mass Transfer Processes ChE 313 treats separation processes in great detail, Reactor Design ChE 321 studies the engineering concepts in industrial-scale reactors, and Process Design ChE 396/7 investigates the creation of new and better processes. If a student can master the fundamentals in ChE 210, this knowledge will be of benefit throughout the rest of the curriculum as well as in a professional engineering career.

It is assumed that the students in the course have a thorough grounding in elementary mathematics, and this mathematical background will be used; some mathematics review may be necessary at various points in the course. By the end of the term the students should understand material and energy concepts; they should be able to apply control volume analysis to a variety of situations in order to solve elementary chemical engineering process problems.

The analysis of chemical processes is conceptually difficult and requires time and effort on the student's part before it can be well understood. Students should read the appropriate sections of the text before lectures, homework assignments should be started as soon as possible, and thoughtful questions should be brought up during lecture and discussion. If you have actually read the entire syllabus, please send me an email so I can congratulate you. Good luck!